

Phase 01 — Evidence synthesis · Pictorial walkthrough

Phase context. Before any new study runs, the team needs to know what the existing evidence base looks like — for the protocol's background, the IRB packet's "what's known", the grant's justification of need, and for guideline / HTA submissions whose entire artefact is a synthesis. This phase covers everything from the first search query through the published systematic review.

Captured against a **live instance** (<http://localhost:8000> , AWS Bedrock `claude-haiku-4-5`) on 2026-06-19. Phase 01 has **8 demos (E1–E8)**. This guide gives the **three core, richest demos (E1–E3) full pictorial treatment**; E4–E8 are summarised as triggers + expected output at the end (they reuse the same card/page patterns and, for NMA/IPD/GRADE, deliver their value as downloadable PDF reports).

Setup notes

- Run the stack auth-disabled (blank Cognito) for clean single-user capture — see the Phase 02 guide.
- **E1 (and E4/E5) render their forest/network plots in the Docker sandbox** — build it once: `docker build -t research-assistant-sandbox:latest ./sandbox` .
- The chat UI requires the Babel 7 pin in `index.html` / `sr.html` (see Phase 02 guide's CDN note).
- E2b (search) and E3 (ingest) hit **live PubMed**.

E1 — Meta-analysis on synthetic data (≈ 5 min)

End-to-end meta-analysis with anti-hallucination guardrails: PICO → search → extraction → forest plot, all inside the sandbox. Deterministic because the extraction data is supplied.

Step 1 — New conversation, paste the research question:

Does adding a proton pump inhibitor to dual antiplatelet therapy reduce upper GI bleeding in patients who recently underwent PCI for acute coronary syndrome?

The model returns a structured **PICO** card.

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INTERVENTION

Proton pump inhibitor (PPI) added to dual antiplatelet therapy (DAPT)

COMPARISON

Dual antiplatelet therapy alone (no PPI)

OUTCOMES

Upper gastrointestinal bleeding

Major gastrointestinal bleeding

Clinically significant bleeding

+ Add

INCLUSION CRITERIA

Randomized controlled trials (RCTs)

Studies in adults with acute coronary syndrome post-PCI

Intervention: PPI plus DAPT vs. DAPT alone

Outcome reporting on gastrointestinal bleeding

+ Add

EXCLUSION CRITERIA

Observational studies or retrospective analyses

Studies in pediatric populations

Studies without specific GI bleeding outcome data

Non-English language publications

+ Add

STUDY TYPES

Randomized Controlled Trial

+ Add

AGE RANGE

≥18 years

NOTES

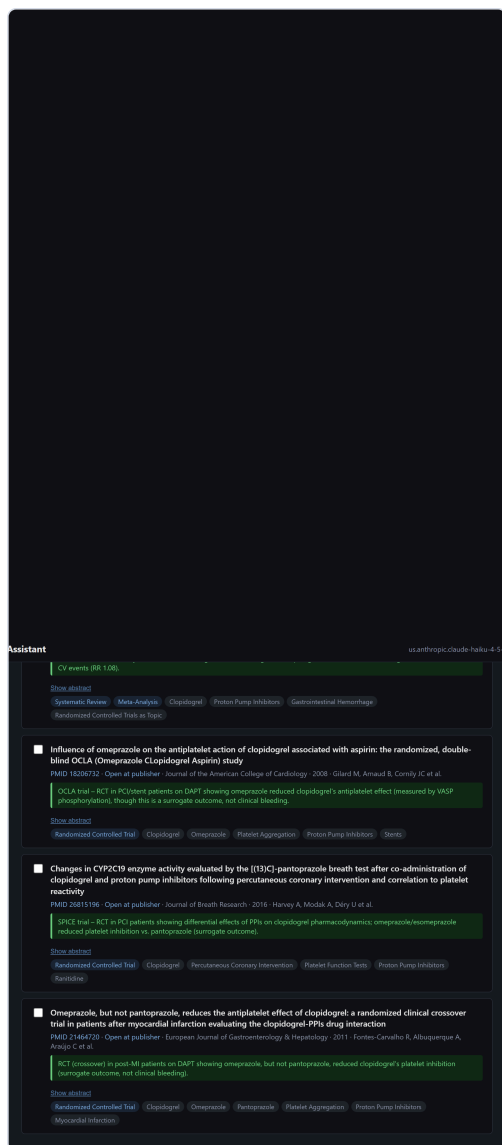
This is a clinically important question as guidelines often recommend gastroprotection with PPIs in patients on DAPT post-PCI. We will search for randomized evidence on whether PPIs truly reduce upper GI bleeding risk compared to DAPT alone. We will focus on hard outcome data (bleeding events) rather than surrogate markers.

The PICO targets the clinical question of gastroprotection efficacy in post-PCI patients. PPI use with DAPT is common clinical practice, but the evidence base should be reviewed. We selected RCTs as the highest quality evidence and focused on GI bleeding outcomes rather than laboratory markers or symptom prevalence. The population is well-defined (ACS post-PCI), and the intervention-comparator distinction is clear (PPI + DAPT vs. DAPT alone)."

Build search strategy

Reset edits

Step 2 — At the PICO step, paste the PICO table → the model runs the search (live PubMed) and returns a **search_results** card. The synthetic studies won't be there — that's expected.



Step 3 — Paste the extraction table (8 fabricated RCTs) → a **data_extraction** card with every row tagged **USER PROVIDED** + **complete**. (Showing 3 of 8 rows.)

Study	Year	Design	PPI events	PPI total	Control events	Control total	Follow-up (mo)
Anderson et al.	2014	Multicenter RCT	12	1200	28	1198	12
Bekele et al.	2016	Single-center RCT	4	320	9	318	6
... (8 studies total)							

Faruq et al. (2021)

PMID · Multicenter RCT · follow-up 12 months

Upper gastrointestinal bleeding

RR (risk ratio) ▼ USER PROVIDED complete

EVENTS (INTERVENTION)

10

N (INTERVENTION)

1400

EVENTS (COMPARISON)

25

N (COMPARISON)

1395

Garcia et al. (2022)

PMID · RCT · follow-up 6 months

Upper gastrointestinal bleeding

RR (risk ratio) ▼ USER PROVIDED complete

EVENTS (INTERVENTION)

3

N (INTERVENTION)

480

EVENTS (COMPARISON)

8

N (COMPARISON)

478

Huang et al. (2023)

PMID · Multicenter RCT · follow-up 12 months

Upper gastrointestinal bleeding

RR (risk ratio) ▼ USER PROVIDED complete

EVENTS (INTERVENTION)

14

N (INTERVENTION)

1800

EVENTS (COMPARISON)

33

N (COMPARISON)

1810

Run meta-analysis (8/8 rows complete)

Assess risk of bias

Reset edits

Step 4 — Click "Run meta-analysis (8/8 rows complete)" → the model calls `sandbox_exec`, which renders the forest plot and pools the estimate.

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Egger's test suggests potential publication bias ($p=0.0016$), but this likely reflects the fact that smaller studies were less likely to show benefit (or were not published), whereas the large, definitive trials consistently demonstrated harm reduction. No study reported cardiovascular complications from PPI use; the evidence base strongly supports gastroprotection with PPI in post-PCI ACS patients receiving DAPT. Caveats: 1 Publication bias concern: Egger's test $p=0.0016$ suggests possible small-study effects; however, large multicenter trials drove the pooled estimate and consistently showed benefit. 2 Surrogate outcome in some studies: While primary endpoints were hard clinical events (bleeding), some trials also reported laboratory markers or symptom-based measures. 3 PPI variability: Meta-analysis pooled different PPIs (omeprazole, esomeprazole, pantoprazole); some have higher CYP2C19 inhibition than others, which may affect clopidogrel efficacy—this was not definitively associated with increased CV events in any trial. 4 Short-term follow-up: Most studies followed patients for 6–12 months; longer-term safety data limited. 5 No reporting of cardiovascular harm: The analysis focused on GI bleeding; subgroup analyses in high-risk populations did not show excess MI or stent thrombosis with PPI co-therapy. 6 Homogeneous population: Results apply specifically to post-PCI ACS patients on DAPT; generalizability to other populations (e.g., chronic stable CAD on clopidogrel alone) not addressed.]

Upper gastrointestinal bleeding (RR)

0.44 95% CI [0.34, 0.58]

I^2 : 0% heterogeneity p : 0.996 p : 0.000 studies: 8 n : 17,596

Adding a PPI to DAPT significantly reduces the risk of upper GI bleeding by approximately 56% (RR 0.44, 95% CI 0.34–0.58, $p<0.001$) in patients post-PCI for acute coronary syndrome, with no evidence of heterogeneity.

Forest Plot: PPI + DAPT vs. DAPT Alone for Upper GI Bleeding in Patients Post-PCI for Acute Coronary Syndrome

Study	RR (95% CI)
Anderson et al. (2014)	0.44 (0.34, 0.58)
Bekele et al. (2016)	0.44 (0.34, 0.58)
Chen et al. (2018)	0.44 (0.34, 0.58)
Diaz et al. (2019)	0.44 (0.34, 0.58)
Eriksen et al. (2020)	0.44 (0.34, 0.58)
Faruq et al. (2023)	0.44 (0.34, 0.58)
Pooled	0.44 (0.34, 0.58)

Describe your clinical question, or answer the assistant's last question... Send

Step 5 — Download the manuscript-style report ([phase01-evidence/e1-meta-analysis.pdf](#)).

Sanity checks: pooled **RR $\approx$ 0.44 (95% CI 0.34–0.58)**, random-effects, 8 study rows + diamond, $n \approx 17,596$. The "Run meta-analysis" button (not a free-text "go ahead") is what emits the extraction JSON that triggers the sandbox — anti-hallucination by design.

E2 — Real evidence pipeline: Protocol → Search → Meta → RoB

Demonstrates **coherence across handoffs** — the same PICO carries through every specialist.

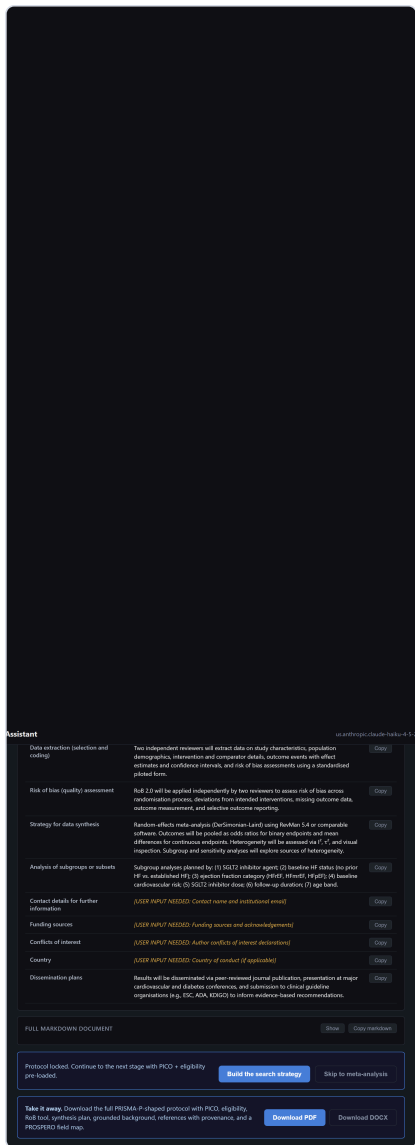
E2a — PRISMA-P protocol drafter

Trigger: `/protocol SGLT2 inhibitors for HF prevention in T2DM`

The `protocol_methods` card lets you edit PICO + eligibility + synthesis plan before drafting.

[illegible]

Click **Methods confirmed — draft full protocol** → a grounded `protocol_document` with numbered citations + PROSPERO field map → **Finalize protocol** → final card with handoff buttons + PDF/DOCX.



E2b — Search strategy

Trigger: /search proton pump inhibitor + DAPT in post-PCI ACS

A `query_blocks` card with MeSH chips — **Confirm terms — compose & test** → a `strategy_result` card with the composed Boolean query and **live PubMed hit counts**.

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("Dual Anti-Platelet Therapy"[MeSH] OR "DAPT"[tiab] OR "dual antiplatelet"[tiab] OR "clopidogrel"[tiab] OR "ticagrelor"[tiab] OR "prasugrel"[tiab])

Drug interaction, efficacy, safety, cardiovascular events, or clinical outcomes

drug interaction

```
("drug interaction"[tiab] OR "drug-drug interaction"[tiab] OR "efficacy"[tiab] OR "safety"[tiab] OR "cardiovascular event"[tiab] OR "PACE"[tiab] OR "stent thrombosis"[tiab] OR "bleeding"[tiab] OR "hemorrhage"[tiab])
```

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Requires institutional Endnote subscription

cochrane central

Cochrane CENTRAL Boolean

not executed here — copy & run manually

Copy

(MeSH: 'Acute Coronary Syndrome' OR MeSH: 'Percutaneous Coronary Intervention' OR ('post-PCI' OR 'after PCI')) AND (MeSH: 'Percutaneous Coronary Intervention' OR 'PCI' OR 'stent') AND (MeSH: 'Proton Pump Inhibitors' OR 'PPI' OR 'PPIs' OR 'omeprazole' OR 'esomeprazole' OR 'lansoprazole' OR 'pantoprazole' OR 'rabeprazole') AND (MeSH: 'Dual Antiplatelet Therapy' OR 'DAPT' OR 'dual antiplatelet' OR 'clopidogrel' OR 'ticagrelor' OR 'prasugrel') AND ('drug interaction' OR 'drug-drug interaction' OR 'efficacy' OR 'safety' OR 'cardiovascular event' OR 'MACE' OR 'stent thrombosis' OR 'bleeding' OR 'hemorrhage')

Publicly accessible

SAMPLE HITS (5)

Comparison of the Effect of Proton Pump Inhibitors Versus Histamine-2 Receptor Antagonists on Cardiovascular Outcomes on Clopidogrel-Based Dual Antiplatelet Therapy: A Nationwide Cohort Study.
PMID 41324390 · The Annals of pharmacotherapy · 2026 · pubmed

Role of Proton Pump Inhibitors in Prevention of Upper Gastrointestinal Bleeding in Patients on Dual Antiplatelet Therapy: Systematic Review and Meta-Analysis.
PMID 40138683 · American journal of therapeutics · 2025 · pubmed

Effect of Potassium-Competitive Acid Blockers on Upper Gastrointestinal Bleeding in Patients on Dual Antiplatelet Therapy After Percutaneous Coronary Intervention: A Nationwide Cohort Study.
PMID 40289422 · Journal of gastroenterology and hepatology · 2025 · pubmed

Analysis of risk factors for gastrointestinal bleeding in percutaneous coronary intervention patients treated with dual antiplatelet therapy after surgery.
PMID 40193633 · Medicine · 2025 · pubmed

Risk Factors for Upper Gastrointestinal Bleeding in Patients Undergoing Percutaneous Coronary Intervention on Dual Antiplatelet Therapy with Assessment of Anti-Ulcer Medication Effects.
PMID 40123608 · Acta Cardiologica Sinica · 2025 · pubmed

Finalize strategy

Query executed successfully with 110 PubMed hits, well within the target band of 50–500. The search captured recent high-quality studies including systematic reviews, meta-analyses, and registry-based cohort studies examining the interaction between PPIs and DAPT in post-PCI ACS patients. Results span from 2016 to 2026, providing comprehensive coverage of the evidence landscape."antmlparameter> <parameter name="refinement_suggestions">[]

E2c — Hand off to meta-analysis (narrated)

The finalized strategy card carries a **Run meta-analysis on these results** button; the data-extraction card carries **Assess risk of bias**. Each seeds a new thread that arrives at the right step with the prior context — the PICO never gets re-typed. (Drop in E1's synthetic extraction for a deterministic forest plot.)

E2d — Risk of bias

Trigger: /rob Run RoB 2.0 on PMID 20925534, PMID 30873575, PMID 31091374

A `rob_assessments` card with the RoB 2.0 domains per study (judgments editable, with verbatim quote justifications pulled from the abstract).

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Bias in measurement of the outcome

Show justification

No information

The abstract does not provide details about validation of measurement instruments or blinding of outcome assessors.

Alpelisib for PIK3CA-Mutated, Hormone Receptor-Positive Advanced Breast Cancer

OVERALL

Some concerns

PMID: 39383124

Randomized controlled trial

Bias arising from the randomization process

Show justification

No information

The abstract describes a randomized phase 3 trial but does not specify the method of sequence generation, allocation, or.

Bias due to deviations from intended interventions

Show justification

No information

The abstract does not specify whether participants or outcome assessors were blinded, or provide information on protocol.

Bias due to missing outcome data

Show justification

No information

The abstract truncates the results section and does not provide information on the number of participants completing the.

Bias in measurement of the outcome

Show justification

No information

The abstract does not describe outcome assessment methods or whether outcome assessors were blinded.

Bias in selection of the reported result

Show justification

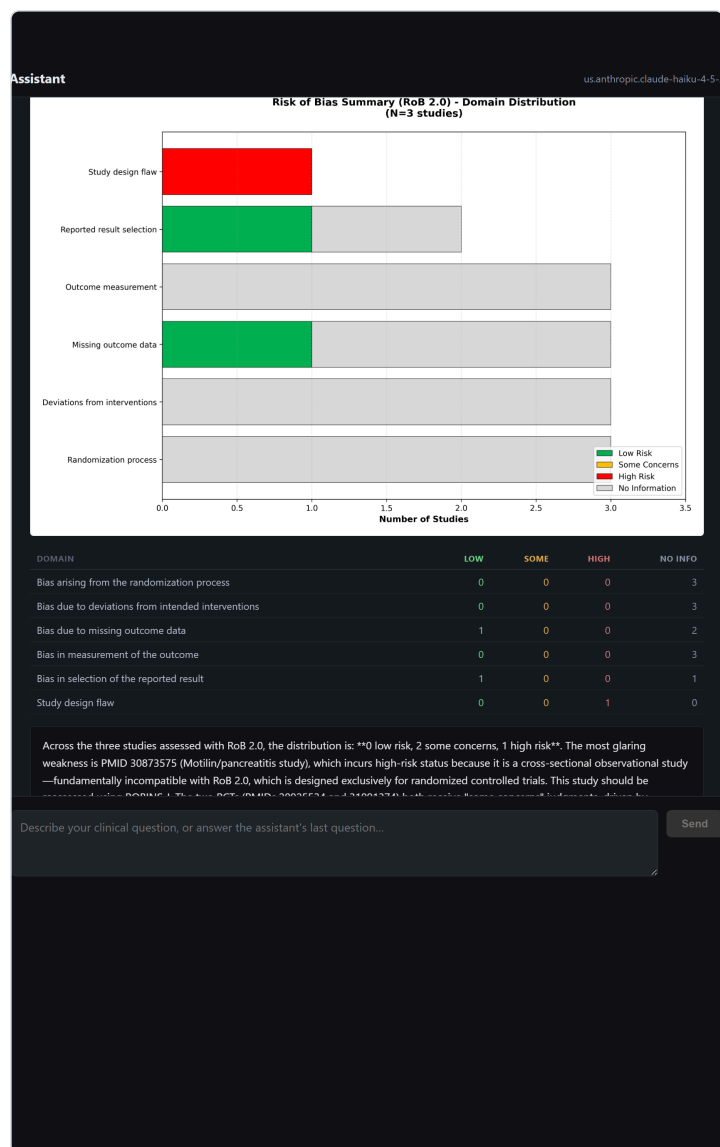
No information

While progression-free survival is mentioned as the primary outcome, the abstract is truncated and does not fully detail.

RoB confirmed — generate summary

Reset edits

Click **RoB confirmed — generate summary** → a stacked-bar `rob_summary` + per-domain counts → **Finalize RoB** → handoff to a sensitivity re-pool + PDF/DOCX.



E3 — SR title/abstract + full-text screening UI

The dual-review screening surface that bridges search (thousands of hits) and meta-analysis (the dozen that qualify). **Open at** <http://localhost:8000/sr.html>.

Step 1 — Create a project (name, PICO, PubMed Boolean query, inclusion/exclusion criteria).

SYSTEMATIC-REVIEW PROJECTS

Title/abstract + full-text dual screening with conflict adjudication, AI-assist, and PRISMA flow diagram. Each project carries its own PICO, search query, inclusion/exclusion criteria, and reviewer roster.

New SR project

NAME

PPI + DAPT upper-GI-bleeding screening (smoke review)

DESCRIPTION

(optional) Plain-language summary of the question

PICO (ONE PER LINE, E.G. "POPULATION: ...")

population: adults post-PCI for ACS on dual antiplatelet therapy
intervention: PPI added to DAPT
comparator: DAPT alone or DAPT + placebo
outcome: upper GI bleeding events

SEARCH QUERY (PUBMED BOOLEAN)

("proton pump inhibitors"[MeSH Terms] OR PPI) AND ("dual antiplatelet"[tiab]
OR DAPT) AND (bleeding OR hemorrhage OR haemorrhage)

INCLUSION CRITERIA (ONE PER LINE)

Adults >=18
Dual antiplatelet therapy
Reports upper GI bleeding

EXCLUSION CRITERIA (ONE PER LINE)

Preclinical / animal
Review articles
Non-English

Create project

Cancel

No SR projects yet — start one above.

Step 2 — Ingest + AI-assist. Click ► **Ingest from search** (live PubMed fan-out → here **52 candidates**) then ✨ **AI-assist** (the `sr_screening_assist` specialist pre-classifies abstracts).

SR Reviews

Back to assistant

OverviewScreen — abstractScreen — full textConflictsPRISMA

PPI + DAPT upper-GI-bleeding screening (smoke review)

ABSTRACT SCREENING

PICO

```
{
  "population": "adults post-PCI for ACS on dual antiplatelet therapy",
  "intervention": "PPI added to DAPT",
  "comparator": "DAPT alone or DAPT + placebo",
  "outcome": "upper GI bleeding events"
}
```

INCLUSION CRITERIA

- Adults >=18
- Dual antiplatelet therapy
- Reports upper GI bleeding

SEARCH QUERY

("proton pump inhibitors"[MeSH Terms] OR PPI) AND ("dual antiplatelet" [tiab] OR DAPT) AND (bleeding OR hemorrhage OR haemorrhage)

EXCLUSION CRITERIA

- Preclinical / animal
- Review articles
- Non-English

Total candidates: 52

Remaining (abstract phase): 52

Decisions logged: 0

Ingest from search

AI-assist (25 abstracts)

Start screening →

Resolve conflicts

Reviewer team

No reviewers assigned yet.

reviewer@example.com

reviewer_1

+ Add

Step 3 — Screen. Each abstract shows an inline **AI-ASSIST** suggestion (include / exclude / maybe + confidence + reasoning); reviewers decide with **i** / **e** / **m** keyboard shortcuts.

← Project **Abstract phase** Full-text phase

Remaining in your queue: **52** Total in project: **52** Shortcuts: **i** include · **e** exclude · **m** maybe

Evaluation of Rabeprazole for Gastrointestinal Protection in Acute Coronary Syndrome Patients Treated with Aspirin and Ticagrelor: A Pilot Study.

Hyun HK, Lee OH, Roh JW, Heo SJ, et al. · Yonsei medical journal · 2026 · PMID 41914314

AI-ASSIST **MAYBE** · confidence 45%

The abstract describes a single-center, prospective observational trial of rabeprazole (a PPI) added to DAPT (aspirin + ticagrelor) in ACS patients post-PCI, matching the population, intervention, and comparator elements. However, the study lacks a control arm—it is described as an "observational trial" with participants receiving rabeprazole, not a randomized comparison of PPI+DAPT versus DAPT alone or placebo. The primary outcome measured gastric mucosal integrity (Lanza score) rather than the pre-specified outcome of upper GI bleeding events; while the abstract implies GI safety, no actual bleeding events are reported. The study population and intervention align with inclusion criteria, but the absence of a comparator group and lack of reported bleeding outcomes leave critical design and outcome specification unclear, requiring human review to determine if this meets the SR's requirements.

Patients with acute coronary syndrome (ACS) on ticagrelor-based dual antiplatelet therapy (DAPT) have an increased risk of gastrointestinal (GI) bleeding. However, evidence supporting the protective role of proton pump inhibitors against DAPT-induced gastric mucosal injury remains limited. This was a single-center, prospective, open-label, observational trial enrolling patients with ACS who were treated with DAPT, specifically aspirin and ticagrelor, following percutaneous coronary intervention in South Korea. Participants received 20 mg rabeprazole once daily for 8 weeks. The primary outcome was the proportion of patients exhibiting a modified Lanza score (MLS) of 0-5 on upper endoscopy at 8 weeks compared to baseline endoscopy results. The secondary outcomes included GI symptom scores and safety assessments. Among the 50 patients included in the per-protocol analysis, the median MLS at baseline and 8 weeks was 2.0 (1.0-2.0) and 2.0 (1.0-2.0), respectively, with no significant change (In this pilot study, rabeprazole was associated with maintained gastric mucosal integrity in patients with ACS receiving ticagrelor-based DAPT.

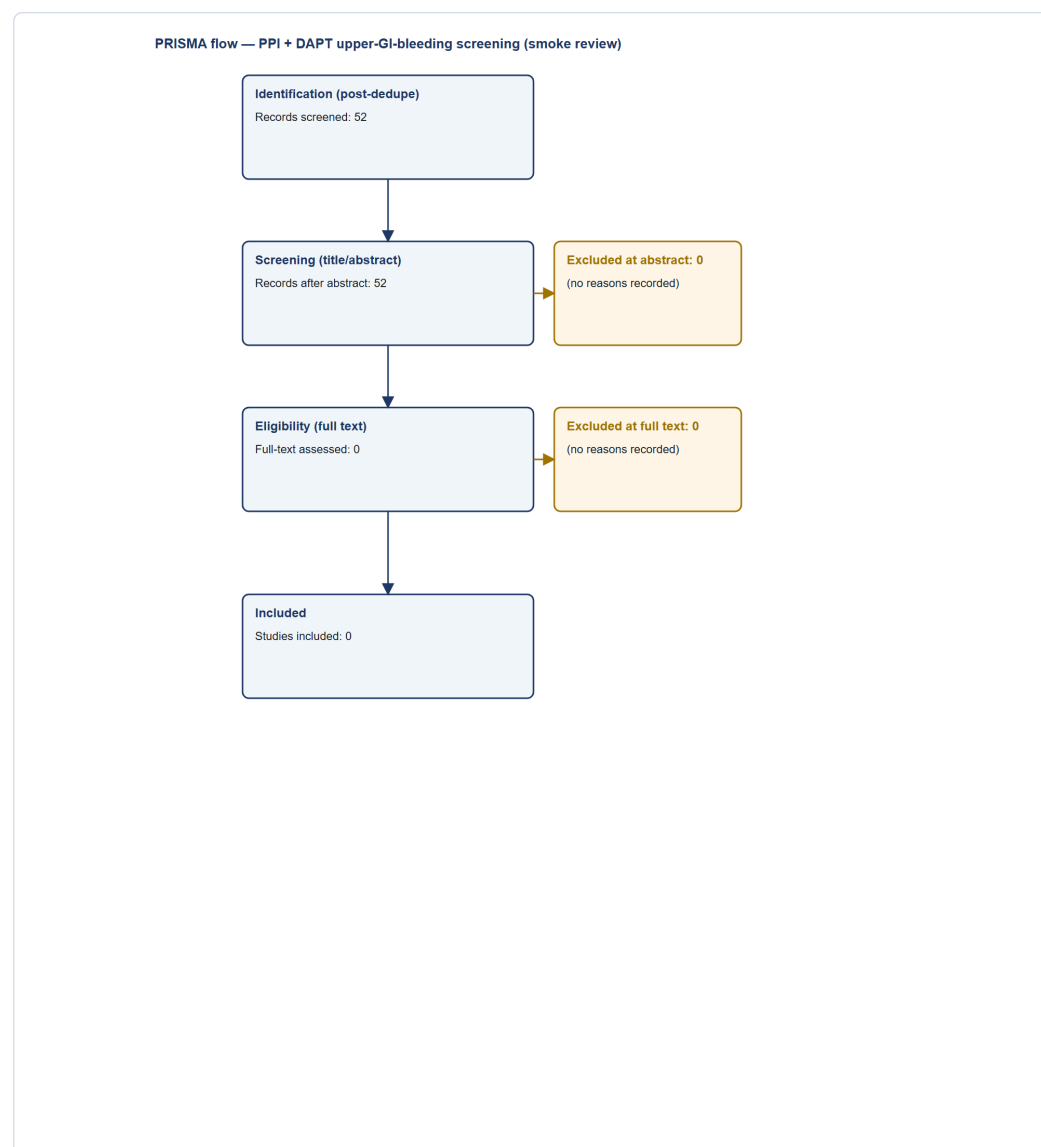
Include (i) **Maybe (m)** **Exclude (e)**

— reason (for exclude) — ▼

Notes (optional)

Dual review (narrated). Reviewer 1 and Reviewer 2 screen **blind** to each other; conflicts surface in the **Conflicts** tab for an adjudicator. This requires separate user accounts, so it isn't shown in this single-user capture — the roster + roles (reviewer_1 / reviewer_2 / adjudicator) are managed in the "Reviewer team" panel on the overview.

Step 4 — PRISMA flow. The **PRISMA** tab renders a hand-rolled SVG with per-stage counts + side exclusion boxes, derived entirely from candidate statuses (no fabrication).



E4–E8 — reference (trigger + expected output)

Per the agreed scope, these five are documented here rather than captured screen-by-screen. E4/E5/E6 are chat workflows whose deliverable is a **downloadable PDF report** (they render as a generic card in chat); E7/E8 live on the dedicated `watches.html` page.

Demo	Trigger / entry	What it produces
E4 — Network meta-analysis	<code>/nma compare five direct oral anticoagulants for stroke prevention in non-valvular AF — apixaban, dabigatran, edoxaban, rivaroxaban, warfarin</code>	5-stage workflow → PicoNetwork (≥3 interventions + transitivity rationale) → per-arm extraction → league table + SUCRA + network-geometry PNG in a landscape PDF. Frequentist by default; type <code>Run Bayesian NMA</code> for the PyMC backend.
E5 — IPD meta-analysis	<code>/ipd individual patient data meta-analysis on statins for primary prevention of cardiovascular events</code>	5-stage → bundle with per-trial CSV pastes → one-stage + two-stage side-by-side (divergence > 0.2 log-scale flags misspecification) → subgroup × treatment interaction → PDF.
E6 — GRADE SoF + PRISMA 2020	<code>/grade Generate a GRADE SoF for the PPI/DAPT meta-analysis</code> (needs a completed E1/E2 meta in-thread, or click → Draft GRADE on the meta card)	Certainty computed (not asserted) from per-domain ratings; every downgrade cites a source number; 42-item PRISMA 2020 checklist; landscape PDF + DOCX.
E7 — Living-review watch	Build a search (E2b) → click Watch this query → set name + schedule + materiality (0.6) → Create watch . View at <code>http://localhost:8000/watches.html</code> .	Scheduled re-search; on each fire it diffs new PMIDs vs baseline, triages materiality, and lights the notification bell. Run now triggers it on demand.
E8 — Group subscriptions (quorum)	<code>watches.html</code> → Group subscriptions → + New subscription (pick a watch, min votes, min fraction) → Manage members (voters / observers) → vote on a run.	Committee-style consensus: quorum = $\text{yes} \geq \max(\text{min_votes}, [\text{min_fraction} \times \text{n_voters}])$; observers excluded from the denominator; on quorum the whole panel is notified. (Multi-voter quorum needs multiple accounts.)

Reproduce

```
uv run python scripts/demo_e1_meta.py # E1 -> phase01-evidence/e1-*
uv run python scripts/demo_e2_pipeline.py # E2 -> phase01-evidence/e2*
```

```
uv run python scripts/demo_e3_sr.py # E3 -> phase01-evidence/e3-*
```